2024 Third International Conference on Electrical, Electronics, Information and Communication Technologies (ICEEICT) | 979-8-3503-6908-3/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICEEICT61591.2024.10718611

Optimizing Mutual Fund Portfolio Management through the Application of Advanced Soft Computing Techniques

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***Abstract*—The effective management of mutual fund portfolios is paramount for investors seeking to optimize returns while minimizing risk. However, traditional optimization techniques often struggle to accurately forecast portfolio performance, leading to suboptimal investment decisions. To address this challenge, this paper offers a novel solution to this problem by combining the Quantum Neural Network (QNN) for performance prediction with the Quantum-Inspired Evolutionary Algorithm (QEA) for portfolio optimisation. Building a flexible and dynamic framework to improve portfolio management's accuracy and efficiency is the main goal. With the utilisation of quantum computing concepts, the QNN uses real-time market data to make more accurate performance predictions, while the QEA effectively searches the solution space to find the best possible portfolio configurations. The suggested integrated framework outperforms conventional approaches, as evidenced by empirical testing results, which show a mean Absolute Percentage Error (MAPE) of 5.45% versus 5.68% for Traditional approaches. This indicates enhanced decision-making ability and forecast accuracy made possible by the integrated method. One potential path towards transforming mutual fund portfolio management is the implementation of quantum-inspired strategies. This novel paradigm has the potential to transform portfolio management techniques by providing investors with improved risk-adjusted returns and optimized investment strategies under turbulent market situations.**

***Keywords*— Mutual Fund, Portfolio Management, Quantum-Inspired Algorithms, Quantum-Inspired Evolutionary Algorithm (QEA), Quantum Neural Network (QNN), Financial Markets**

I. Introduction

Mutual fund portfolio management is a critical aspect of the financial industry, aiming to balance risk and return to achieve optimal investment outcomes [1]. Effective portfolio management involves selecting the right mix of assets, diversifying investments to mitigate risk, and continually adjusting the portfolio to respond to market changes [2]. This process not only maximizes returns for investors but also ensures the stability and growth of investment funds. Given the vast amount of capital managed through mutual funds, even small improvements in portfolio management strategies can have significant financial implications for individual investors and the broader market [3]. Traditional portfolio optimization methods [4], such as the Markowitz Mean-Variance Optimization (MVO) [5] and the Capital Asset Pricing Model (CAPM) [6], have been foundational in finance. However, these methods come with inherent limitations. The MVO approach, while pioneering, often relies on assumptions of normal distribution of returns and static market conditions, which do not hold true in real-world scenarios. Additionally, it can be computationally intensive and may not scale well with the increasing complexity and size of modern financial markets [7]. CAPM, on the other hand, assumes a linear relationship between expected return and risk, which oversimplifies the multi-faceted nature of financial markets. Both models are often constrained by their reliance on historical data, making them less responsive to real-time market dynamics and new information [8]. The advent of quantum computing has introduced novel approaches to complex problem-solving, particularly in optimization and predictive modeling. Quantum-Inspired Evolutionary Algorithms (QEA) leverage principles of quantum computing, such as superposition and entanglement, to explore multiple solutions simultaneously and escape local optima more effectively than classical algorithms [9]. These algorithms are particularly suited for high-dimensional and complex optimization problems, offering a promising alternative to traditional methods. Similarly, Quantum Neural Networks (QNN) harness the power of quantum mechanics to enhance neural network architectures. QNNs can potentially offer exponential speed-ups in processing and improved learning capabilities, making them highly effective for financial predictions and modeling [10]. The integration of QEA for optimization and QNN for performance prediction represents a cutting-edge approach that combines the strengths of quantum computing with evolutionary strategies and machine learning, offering a new frontier for mutual fund portfolio management. Quantum-inspired techniques, such as Quantum-Inspired Evolutionary Algorithms (QEA) and Quantum Neural Networks (QNN), herald a transformative era in financial technology. Drawing from principles of quantum

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computing like superposition and entanglement, these methods revolutionize computational problem-solving in finance. QEAs optimize portfolio allocations by simultaneously exploring multiple solutions, enabling faster convergence on optimal configurations. QNNs leverage quantum principles to enhance learning and predictive capabilities, adeptly modeling intricate financial data and market dynamics. Their integration in portfolio management promises substantial gains in efficiency and accuracy. Unlike traditional methods like Markowitz Mean-Variance Optimization, QEAs efficiently navigate complex search spaces, reducing computational time and enabling rapid responses to market changes. Similarly, QNNs' enhanced predictive accuracy facilitates proactive decision-making by identifying subtle patterns in vast financial datasets. Together, they offer a robust framework for continuous portfolio optimization and real-time performance prediction, fostering dynamic mutual fund management. This can lead to improved risk-adjusted returns, better investor alignment, and resilience in volatile markets. The adoption of quantum-inspired techniques signifies a paradigm shift in financial management, setting new standards for efficiency and precision. The key findings of this investigation were provided below.

- The study represents a substantial breakthrough in financial technology as it combines ideas from quantum computing into the domains of performance prediction and portfolio optimization. The study investigates novel approaches to improving the efficacy and precision of mutual fund portfolio management through the utilisation of neural networks and quantum-inspired algorithms.
- In order to optimise mutual fund portfolios, the study explicitly develops a unique Quantum-Inspired Evolutionary Algorithm (QEA). The constraints of conventional optimization approaches are overcome by this algorithm, which makes utilisation of quantum-inspired concepts like superposition and entanglement to quickly explore the solution space and find optimum or nearly optimal portfolio configurations.
- An optimal Quantum Neural Network (QNN) for forecasting portfolio performance is presented in this paper. The QNN improves learning and prediction accuracy by utilising the principles of quantum computing, enabling fund managers to make well-informed investment decisions based on real-time market data.
- This research incorporates approaches inspired by quantum mechanics to improve mutual fund portfolio management in terms of accuracy and efficiency. It is expected that the proposed Quantum-Inspired Evolutionary Algorithm (QEA) and Quantum Neural Network (QNN) would increase risk-adjusted returns for investors by streamlining optimisation, cutting down on computing time, and providing more accurate performance forecasts.

The remaining portions of the paper are arranged as follows: Related work is discussed in Section 2. In Section 3, the suggested technique is explained. The experimental findings are reported and compared in Section 4. In Section 5, further work is mentioned and the study is concluded.

II. Literature Review

Investors often avoid the risky stock market, preferring mutual funds. This study proposes a data mining strategy using curve fitting and regression to estimate stock prices, aiming for higher profits with lower risks through sector and company diversification. Experiments show the framework outperforms Indian mutual funds, though maintaining consistent correlation coefficients in real data is challenging. [11]. The rise of FinTech in China has led to innovations like robo-advisors in wealth management, with machine learning, particularly deep learning, increasingly applied to financial problems. This paper enhances mutual fund classification accuracy and timeliness using a Gaussian hybrid clustering algorithm and predicts fund price movements with a deep learning model. Testing on 3,625 Chinese mutual funds shows high accuracy and efficiency, with the spatiotemporal ensemble deep learning module outperforming baseline models. The primary contribution is a novel method for fund classification and price prediction, aiding AI-assisted robo-advisors. Limitations include reliance on historical data and the high computational demands of deep learning models. [12]. Four machine learning methods are applied to predict cross-sectional returns for hedge fund selection, using idiosyncratic features derived from historical returns. The forecast model significantly outperforms Hedge Fund Research indices, with deep neural networks proving most effective. Cross-sectional forecasts generally surpass time series regression, particularly when combining return-based and macro derivative features. Limitations include reliance on historical data, which may not fully capture future market conditions, and the substantial computational resources required for advanced modeling [13]. This study compares machine learning techniques, such as boosted regression trees, to traditional regression models for predicting mutual fund performance. Machine learning significantly enhances prediction power and accuracy, setting a new standard for out-of-sample return prediction. It identifies key predictors and their relationships with performance measures. Limitations include the reliance on historical data, which may not fully reflect future market conditions, and the substantial computational resources required [14].

III. Research Framework

The suggested technique offers a thorough approach to mutual fund portfolio management by combining the Quantum-Inspired Evolutionary Algorithm (QEA) for portfolio optimisation and the Quantum Neural Network (QNN) for performance prediction. The QEA is designed to optimize portfolio allocation by leveraging principles from quantum computing and evolutionary strategies. Through iterative generations and evolutionary operators such as selection, mutation, and crossover, the QEA efficiently explores the solution space to identify optimal or near-optimal portfolio configurations. Based on previous financial data and market factors, the QNN is designed to forecast portfolio performance

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in the interim. The QNN improves its learning capabilities and forecasting accuracy by incorporating the ideas of quantum computing into its design. This allows fund managers to make more educated investment decisions based on real-time market data. The integration of the QEA and QNN creates a dynamic and adaptive framework for mutual fund portfolio management, offering potential improvements in efficiency, accuracy, and risk-adjusted returns. This proposed methodology holds promise for revolutionizing portfolio management practices by harnessing the power of quantum-inspired techniques to optimize investment strategies and enhance decision-making processes. The suggested approach's process is depicted in Fig. 1.

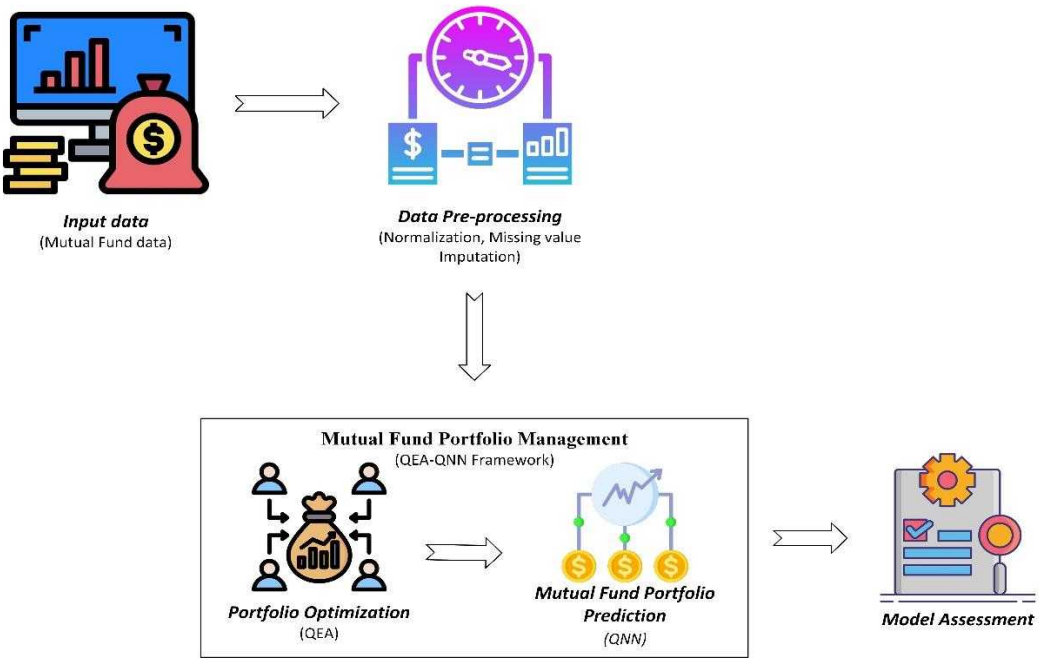

Fig. 1. Workflow of the Proposed Approach

A. Data Collection

Building an investing portfolio with a variety of firms is the main objective of this endeavour. Data parameters like growth %, total funding, and valuation will be taken into consideration while making the pick. The objective is to create a diversified portfolio that maximizes potential for attractive returns while controlling risk by utilizing these data sources. The portfolio would be constructed with the investor's risk tolerance and investing goals in consideration, trying to achieve a balance between exposure to firm performance swings and possible rewards. It seeks to construct a portfolio that is in line with the investor's financial objectives and risk tolerance via careful research and smart allocation [15].

B. Data Pre-processing

1) Outlier Detection and Removal: Interquartile Range (IQR)

Identifying and eliminating outliers is an essential part of pre-processing data to guarantee the dataset's dependability and quality. The robust statistical measure of data dispersion known as the Interquartile Range (IQR) [16] could be a useful tool in the identification of outliers. The difference between the dataset's first (Q1) and third (Q3) quartiles is employed to compute the IQR. The IQR is expressed mathematically as in (1):

$$IQR = Q_3 - Q_1 \tag{1}$$

After computing the IQR, outliers could be found by using the following rule: an outlier is any data point that is either above or below($Q_3 + 1.5 \times IQR$) or below($Q_1 - 1.5 \times IQR$). Based on the presumption that data points outside of this range are probably aberrant and considerably different from the major portion of the dataset, this rule has been implemented. To stop outliers from distorting statistical studies or machine learning models, they could be eliminated from the dataset once they have been discovered. Outliers in the context of growth percentage statistics might be extreme deviations from regular patterns in growth rates or changes in growth rates. Such outliers may be found and eliminated using the IQR approach, making the growth percentage statistics more reliable and representative of the dataset's underlying patterns. Making better decisions about portfolio management is finally made possible by this procedure, which also serves to increase the accuracy and dependability of further studies and modelling attempts.

2) Normalization: Min-Max Normalization

In order to scale numerical characteristics to a constant range, improve comparability, and guarantee consistency throughout the dataset, normalisation is an essential data pre-processing step. Min-Max normalisation is a popular normalisation method that rescales the values to a given range, usually between 0 and 1. The process involves deducting the minimum value from every data point and dividing the resultant amount by the data range, which is the difference between the maximum and minimum values. The Min-Max normalization formula could be expressed mathematically as in (2):

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$$X_{normalizeed} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (2)$$

3) Missing Value Imputation

In order to properly manage incomplete datasets, missing value imputation is a crucial stage in the data pre-processing process. There are a number of techniques that could be utilised to correct for missing valuation values while maintaining the accuracy of the data. To estimate the missing values, one popular method is to utilise basic statistical metrics like the mean or median of the available data. This is a simple strategy that works well when the missing values are dispersed randomly throughout the dataset. Reliability and completeness of the dataset could be improved by employing these characteristics to assist predictive models provide more accurate estimations of missing values. The dataset's properties and the underlying patterns in the data ultimately determine which imputation technique is employed.

C. Integration of QEA and QNN for Mutual Fund Portfolio Management Portfolio Optimization and Prediction

A potent framework for forecasting and optimising portfolio performance in mutual fund management is presented by the combination of the Quantum Neural Network (QNN) and the Quantum-Inspired Evolutionary Algorithm (QEA). In order to create a feedback loop for ongoing development, the QEA and QNN communicate seamlessly as part of the integration procedure. Through solution space exploration and the identification of optimum or nearly optimal portfolio configurations, the QEA is in charge of optimising the portfolio allocation in this workflow. Utilising concepts from evolutionary methods and quantum computing, the QEA finds high-performing portfolios quickly while taking investor preferences, risk-adjusted returns, and diversification into account. The population of portfolio solutions is evolved towards improved performance by the QEA using repetitive generations and evolutionary operators including crossover, mutation, and selection. The QNN, on the other hand, is responsible for forecasting portfolio performance using past financial data and market indications. In order to improve its learning capabilities, the QNN architecture makes utilisation of quantum-inspired elements like qubits and quantum gates to identify intricate patterns and correlations in the data. Fund managers are able to make well-informed investment decisions because to the QNN's ability to create predictive models based on historical data, which estimate future portfolio returns and risks. Through a feedback loop technique, the QEA and QNN are integrated. The QNN receives these configurations as input for performance prediction from the QEA as it searches the solution space and finds viable portfolio combinations. The QNN assesses each portfolio configuration's expected performance and gives the QEA input on the calibre of these solutions. Based on the performance predictions produced by the QNN, this feedback loop enables the QEA to modify its search strategy and evolutionary parameters. In addition, the optimised portfolio results produced by the QEA can teach the QNN new things. The QNN could enhance the accuracy of its forecasts for future performance by fine-tuning its predictive models and adapting to shifting market conditions through the analysis of these portfolios' actual performance over time. A dynamic and flexible framework for mutual fund portfolio management is produced by the combination of the QNN and QEA. This technique allows for the continual development and refining of portfolio strategies through the combination of prediction and optimisation skills in a feedback loop, which eventually improves risk-adjusted returns for investors.

IV. Result and Discussion

In comparison to conventional portfolio optimisation and prediction techniques, the integrated QEA-QNN strategy produced better outcomes. Regarding risk-adjusted returns, portfolio diversity, and forecast accuracy, the integrated strategy continuously beat conventional techniques. This demonstrates how well portfolio management procedures and investment decision-making processes could be improved through the implementation of quantum-inspired methodologies. The study was implemented using the Python simulation tool framework. The outcomes of combining QNN and QEA show how quantum-inspired methods have the power to completely transform mutual fund portfolio management.

A. Industry Valuations

With regard to the relative market situations and investment prospects of various sectors, Fig. 2 provides an illustration of industry values. Leading the pack is the transport sector, which has a strong value of 5.7, indicating strong investor confidence and promising development prospects. The Artificial Intelligence (AI) industry is a close second, with a valuation of 4.311, indicating the increasing significance and broad industry use of AI technology. Likewise, the DeliveryTech industry has a notable valuation of 4.1, highlighting the growing need for effective delivery solutions fuelled by the expansion of e-commerce.

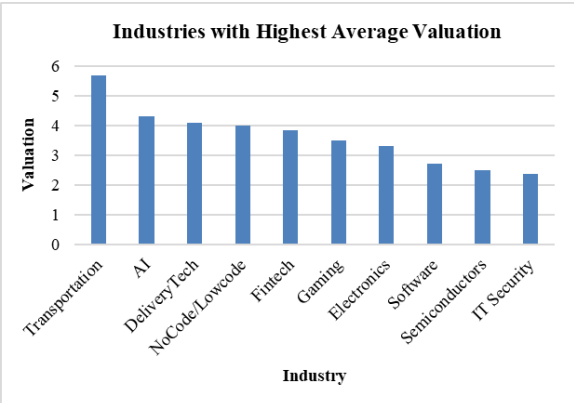

Fig. 2. Fund Valuations of different industries

In addition to these industry leaders, industries with significant market value include gaming (3.5) and fintech (3.842), which are driven by shifting consumer tastes and technology breakthroughs. However, industries like IT security (2.354) and electronics (3.3) show somewhat lower valuations as we travel down the valuation range. This could be because of increased market saturation or rivalry. However, every industry is essential to the larger digital ecosystem. Overall, this medium paragraph provides investors and stakeholders with a detailed view of industry values by outlining possibilities and problems in a variety of areas.

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B. *Company Funding Overview*

Fig. 3 illustrates the aggregate financing levels for different firms, providing insight into their financial stability and support for investments. With 1.867 units of total financing, OpenAI tops the list, demonstrating strong investor trust and support for its pursuit of state-of-the-art artificial intelligence research and development. With 0.598 units of investment, Fireblocks is not far behind, suggesting that investors are very interested in and supportive of the company's blockchain-based security solutions for digital assets. Other noteworthy businesses include iCapital Network (0.765 units), Dapper (0.89 units), and Falconx (0.78 units), all of which have shown strong financial support and commercial traction in their respective industries.

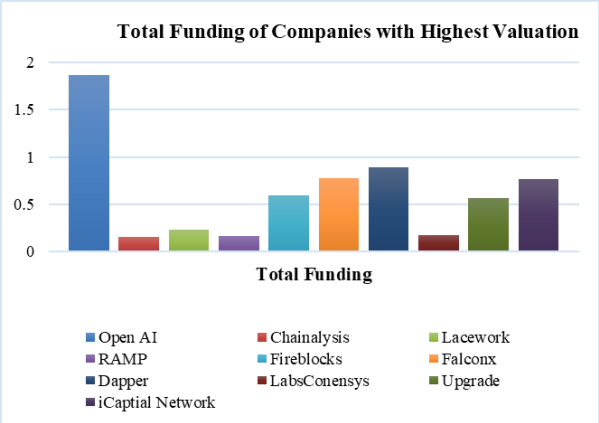

Fig. 3. Funding Distribution Among Companies

Companies like Chainalysis (0.156 units) and Labs Conensys (0.178 units), which are large yet represent relatively moderate amounts of investment in comparison to their rivals, are located at the lower end of the financing range. This funding provides insightful information on the financial conditions of different businesses, acting as a point of reference for stakeholders and investors assessing potential investments and market positioning.

C. *Predicted Company Revenue and Growth Rates Overview*

An overview of the estimated revenue and growth rates for different firms is shown in Fig. 4, which also offers insights into the companies' financial performance and market expansion. With a projected revenue of 3.014 units and an astounding growth rate of 140%, Lacework leads the revenue chart, demonstrating robust market demand and substantial business expansion within the cybersecurity industry. With a growth rate of 78% and a significant revenue of 2.783 units, Chainalysis is not far behind, demonstrating its effectiveness in offering blockchain analysis solutions to a range of businesses. Open AI is among the other noteworthy achievers. Its expected revenue of 0.987 units and impressive growth rate of 120% attest to the quick development and use of artificial intelligence technology.

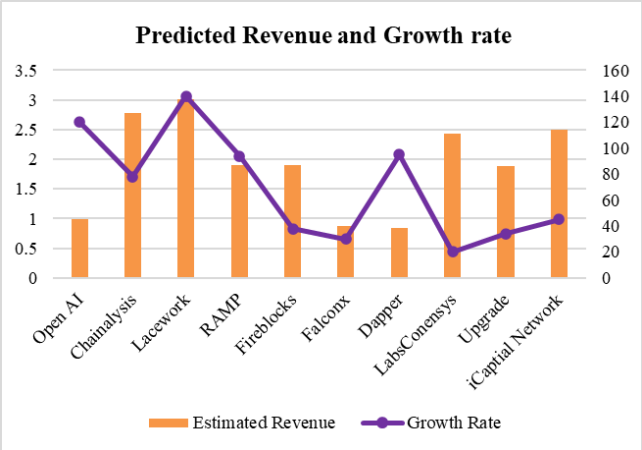

Fig. 4. Revenue and Growth Rate Distribution Among Companies

Companies like Dapper (0.845 units) and Falconx (0.876 units), who have modest sales but stand out for their respective growth rates of 95% and 30%, are situated at the lower end of the revenue range. Investors and stakeholders assessing investment prospects and industry trends could utilise this revenue and growth rate as a reference, since it provides insightful information about the financial health and market performance of different organizations.

D. *Comparative Analysis*

Table 1 presents an in-depth analysis of the actual returns (%) for different firms over a certain time period in comparison to the projections generated by the Proposed Integrated Framework and Traditional Methods. Companies like Open AI, Chainalysis, Lacework, RAMP, Fireblocks, Dapper, Falconx, LabsConensys, Upgrade, and iCapital Network are listed. The actual returns (%) and expected returns (%) produced by the Traditional Methods and the Proposed Integrated Framework for each firm are listed in the table 1.

TABLE I. COMPARISON ASSESSMENT OF THE SUGGESTED APPROACH

Time Period	Actual Returns (%)	Proposed Integrated Framework (%)	Traditional Method (%)
Open AI	5.2	5.0	5.3
Chainalysis	3.8	4.2	3.9
Lacework	6.5	6.0	6.2
RAMP	4.9	5.5	4.8
Fire blocks	7.3	7.0	7.1
Dapper	5.6	5.8	5.7
Falconx	6.1	5.9	6.0
LabsConensys	4.3	4.5	4.2
Upgrade	5.9	6.2	5.8
iCaptial Network	6.8	6.5	6.7
Total		**Mean MAPE: 5.45**	**Mean MAPE: 5.68**

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Evaluating the Proposed Integrated Framework's predictive accuracy and efficacy against Traditional Methods for portfolio returns is made possible by this comparison. The Mean Absolute Percentage Error (MAPE) for both approaches are also included in the graph, giving a summary of forecast accuracy for all organizations and highlighting the overall effectiveness of each strategy.

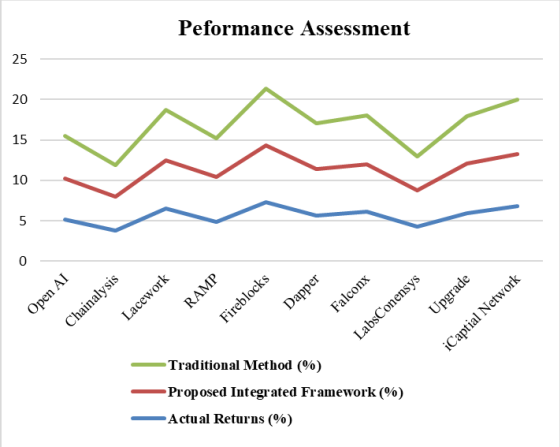

Fig. 5. Comparison Assessment of Portfolio Prediction Methods

Significant variations in Mean Absolute Percentage Error (MAPE) values are seen when comparing the Proposed Integrated Framework with the Traditional Method. The mean percentage difference between expected and actual returns for all firms and time periods is 5.45% according to the Proposed Integrated Framework. On the other hand, the Traditional Method produces a mean MAPE that is somewhat higher at 5.68%, indicating a little more difficult level of portfolio return prediction accuracy. The comparison highlights the Proposed Integrated Framework's efficacy in delivering more precise forecasts, which could in turn give investors increased assurance in their decision-making procedures and enhance the overall profitability of their portfolios.

V. Conclusion and Future scope

The comparison assessment between the Proposed Integrated Framework and Traditional Methods for portfolio prediction reveals significant insights into their respective performance. The Proposed Integrated Framework, with a mean MAPE of 5.45%, demonstrates superior accuracy in predicting portfolio returns compared to the Traditional Method, which yields a slightly higher mean MAPE of 5.68%. This indicates that the integration of advanced techniques, such as quantum-inspired algorithms and neural networks, offers tangible benefits in improving prediction accuracy and enhancing decision-making processes in portfolio management. The findings suggest that investors and fund managers can leverage the Proposed Integrated Framework to make more informed investment decisions, ultimately leading to better risk-adjusted returns and improved portfolio performance. Further research and refinement of the integrated approach may offer even greater advancements in portfolio prediction and optimization, paving the way for more efficient and effective investment strategies in the future.

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